UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 3: Research Protocol Development

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1. Title

Proposed title

Reference Architecture Based on Specialized Generative AI Agents for the Governed Development and Conformance Verification of Critical State Information Systems: An Application Case in the Peruvian Pension System.

Short title

Reference architecture with governed generative AI agents for critical State information systems.

Classification / type of study

Applied research in Design Science Research (DSR), with a mixed (quantitative and qualitative) evaluation strategy embedded in the design cycle. This is neither a pure experimental design nor a purely interpretivist study; it is classified as artifact design-and-evaluation research (Hevner et al., 2004; Peffers et al., 2007).

2. Abstract

Peruvian public entities are increasingly incorporating generative artificial intelligence (AI) into the development of critical information systems, but this integration remains largely unstructured and weakly governed, raising the risk of regulatory non-compliance and failures in essential services. Following a Design Science Research (DSR) approach, this study designs and evaluates a reference architecture that incorporates specialized generative AI agents as governed components of the development process, explicitly separating the generation and verification functions through a conformance principle. The artifact is restricted to the development and verification phases, not to production execution, and its constraints are anchored in Law No. 29733 on the Protection of Personal Data, Law No. 31814 promoting the use of artificial intelligence, and the NTP-ISO/IEC 42001:2025 standard on AI management systems. The artifact will be evaluated through a functional prototype applied to the context of the Peruvian pension system, comparing the performance of a two-agent pipeline (generator and independent, at least partially deterministic, verifier) against a single-agent generation baseline, using as metrics the precision and recall of conformance-violation detection, the defect escape rate, rework effort, and lead time. The evaluation will be complemented by a scenario-based architecture exercise with expert participation. The results are expected to yield a reference architecture and transferable design principles for the governed development of information systems in the public sector, with potential dissemination in venues such as IEEE Software, Journal of Systems and Software, Government Information Quarterly, DESRIST, and ECSA.

3. Introduction and Problem Statement

Context

Peruvian public entities use information systems to support their mission-critical and support processes, and they increasingly incorporate artificial intelligence (AI) into their development, even though this integration remains largely unstructured and weakly governed. There are no systematic mechanisms to verify that code and artifacts generated by AI comply with the enterprise architecture and applicable regulations before advancing through the software lifecycle.

Problem statement

In information systems that support essential State services — such as the Peruvian pension system, this lack of governance over the use of generative AI in development constitutes a concrete risk to the quality and timeliness of services, the protection of personal data, and operational continuity. The research problem can be formulated as follows: how should the use of generative AI agents within the software development cycle be structured so that their conformance with the enterprise architecture and applicable regulations can be verified systematically, and as deterministically as possible, without compromising the continuity of public services or the confidentiality of the sensitive information they process?

Rationale

- Practical relevance: the reliance on critical State systems demands assurance mechanisms that do not yet exist in standardized form for AI-assisted development.
- Theoretical relevance: a reference architecture is needed, not merely a one-off implementation, that documents design principles transferable to other public-administration contexts.
- Regulatory relevance: Law No. 31814 and NTP-ISO/IEC 42001:2025 establish an AI governance framework that has yet to be translated into concrete software-development architectures in the Peruvian public sector.

Scope and delimitation

The artifact operates exclusively in the development and verification phases of the software lifecycle, not in production execution. The agents will run preferably in local, controlled environments, with access restricted to only the strictly necessary code and data, at all times restricting the use of identifiable personal data from the pension system.

4. Literature Review

The preliminary review is organized into five thematic clusters that frame the contribution of this research. This initial synthesis will be expanded during the problem-identification phase of the DSR cycle (Section 6) through a formal systematic or scoping review.

4.1 Enterprise architecture and reference architectures in the public sector

The enterprise-architecture literature provides the governance principles and architectural views that allow AI agents to be treated as governed components, rather than as black boxes external to the system, within a reference architecture for State entities. It is important to highlight the alignment of this enterprise architecture with Supreme Decree No. 029-2021-PCM, which approves the Regulations of Legislative Decree No. 1412, the Digital Government Law, establishing provisions on the conditions, requirements, and use of technologies and electronic means in administrative procedures, particularly under TITLE VIII, DIGITAL ARCHITECTURE.

4.2 Generative AI agents in software engineering

A growing body of work explores the use of generative models for code generation and development artifacts, as well as multi-agent architectures in which distinct specialized agents fulfill differentiated roles within the development cycle.

4.3 Conformance verification, static analysis, and policy-as-code

Static-analysis and policy-as-code techniques provide the foundation for a verification mechanism that does not depend solely on one generative model detecting another's errors, thereby mitigating the correlation of failures between generator and verifier.

4.4 AI governance and the Peruvian regulatory framework

Law No. 29733 (Protection of Personal Data), Law No. 31814 (promoting the use of artificial intelligence), and NTP-ISO/IEC 42001:2025 (AI management systems) constitute the regulatory framework that anchors the conformance requirements of the proposed artifact. Supreme Decree No. 029-2021-PCM, which approves the Regulations of Legislative Decree No. 1412, the Digital Government Law, TITLE VIII, DIGITAL ARCHITECTURE, is also part of this framework.

4.5 Methodological foundations of Design Science Research

The work of Hevner et al. (2004), Peffers et al. (2007), and Gregor and Hevner (2013) underpins the paradigmatic choice (documented in the Paradigm Justification Statement deliverable) and the six-activity methodology adopted in Section 6 of this protocol.

Research gap

The preliminary review did not identify a reference architecture that combines (a) specialized, governed generative AI agents, (b) a conformance principle that is at least partially deterministic and independent of the generative process itself, and (c) empirical evaluation in a real case of a critical State information system within a regulated Latin American context. This is the gap that the present research seeks to close.

5. Research Questions, Hypotheses, and Objectives

Main research question

How should a reference architecture based on specialized generative AI agents be structured to automate and verify the development of critical state information systems, ensuring their compliance with the enterprise architecture and applicable regulations?

Secondary questions

1. What architectural principles and constraints must the artifact incorporate to be governable and auditable?
2. What conformance-verification techniques (static analysis, policy as code, rule-checking) reduce the reliance on one generative model detecting another's errors?
3. To what extent does a two-agent pipeline (generator and independent verifier) improve, relative to a single-agent baseline, the detection of conformance violations, the defect escape rate, rework effort, and lead time?
4. How do architecture and domain experts perceive the usefulness, feasibility, and risks of the artifact in a scenario-based evaluation exercise?

Design propositions

Consistent with the DSR approach adopted (see the Paradigm Justification Statement deliverable), this protocol does not formulate hypotheses in the positivist sense, but rather testable design propositions:

— P1: a two-agent pipeline with independent, at least partially deterministic verification will achieve higher precision and recall in detecting conformance violations than single-agent generation without independent verification.

— P2: the two-agent pipeline will reduce the defect escape rate and rework effort relative to the baseline.

— P3: architecture and domain experts will rate the reference architecture as useful and feasible in the scenario-based evaluation, identifying manageable risks.

General objective

To design and evaluate a reference architecture based on specialized generative AI agents, governed by a conformance principle, for the development and verification of critical State information systems, through an application case in the Peruvian pension system.

Specific objectives

1. Systematize the architectural and regulatory requirements (Law No. 29733, Law No. 31814, NTP-ISO/IEC 42001:2025) that the reference architecture must satisfy.
2. Design the artifact, reference architecture and conformance-verification mechanism, following enterprise-architecture principles.
3. Implement a functional prototype of the generation/verification pipeline in a local, controlled environment.
4. Evaluate the prototype in a real application case, comparing it against a single-agent baseline using the metrics defined in Section 6.
5. Conduct a scenario-based architecture evaluation with the participation of architecture and domain experts.
6. Formulate transferable design principles and disseminate the results in relevant academic venues.

6. Methodology

6.1 Selected paradigm and method

Design Science Research (DSR) is adopted, selected in the Method Fit Matrix (Deliverable 2) with a weighted score of 4.65/5 (93/100), against 2.45/5 (Pure Experiment), 2.50/5 (Simulation), and 3.65/5 (Mixed Methods as an independent paradigm). Pure Experiment and Simulation are eliminated by the ≤2 gate rule on a critical criterion (Contribution for both; also Epistemological Fit for Simulation), because neither can produce or govern the reference-architecture artifact that is the intended contribution. Mixed Methods remains a viable evaluation strategy embedded within DSR — not the governing method — consistent with the paradigm justification.

Criterion / Weight	Pure Experiment	Simulation	Mixed Methods	DSR (selected)
Epistemological Fit (30%)	3/5	2/5	4/5	**5/5**
Data Feasibility (20%)	2/5	3/5	3/5	**4/5**
Execution Feasibility (15%)	2/5	3/5	3/5	**4/5**
Contribution (20%)	2/5	2/5	4/5	**5/5**
Venue Fit (15%)	3/5	3/5	4/5	**5/5**
Weighted Total	2.45/5 (49/100)	2.50/5 (50/100)	3.65/5 (73/100)	**4.65/5 (93/100)**

6.2 Methodological cycle (Peffers et al., 2007)

1. Problem identification and motivation: synthesis of the AI-governance gap in the development of critical State systems (Sections 3 and 4).
2. Definition of the objectives of a solution: functional and non-functional requirements of the reference architecture, derived from the enterprise architecture and the regulatory framework.
3. Design and development: specification of the artifact - architectural views, generator/verifier agent roles, and the conformance mechanism - and construction of a functional prototype.
4. Demonstration: deployment of the prototype in the application case (Peruvian pension system), in a local/controlled environment, restricted to development and verification.
5. Evaluation: quantitative comparison against the single-agent baseline and qualitative scenario-based architecture evaluation with experts.
6. Communication: dissemination of results and design principles in the venues identified in Section 8.

6.3 Evaluation design

The evaluation combines two complementary components within the DSR cycle, consistent with the strategy set out in the Method Fit Matrix: a controlled comparison of the two-agent pipeline against a single-agent baseline, and simulation used for pre-deployment stress testing of defect-injection and correlated generator–verifier failure scenarios.

— Controlled pipeline comparison: the two-agent pipeline (generator + independent, at least partially deterministic verifier) is compared against a single-agent generation baseline without independent verification, on the same development tasks from the application case.

— Scenario-based architecture evaluation: a panel of enterprise-architecture and domain experts evaluates the usefulness, feasibility, and risks of the artifact against representative scenarios (e.g., a regulatory change, a conformance incident, scaling to a second case).

6.4 Primary metrics

— Precision and recall in the detection of conformance violations against enterprise-architecture and regulatory rules.

— Defect escape rate (defects not caught by the verifier that reach later stages).

— Rework effort (hours or iterations required to fix detected defects).

— Lead time of the generated and verified software artifact.

6.5 Data sources

Data sources for the selected DSR plan: prototype interaction and build logs; automatically detected conformance violations (precision/recall) against enterprise-architecture and regulatory rules; defect escape rate, rework effort, and lead time; the single-agent (no independent verification) baseline; enterprise-architecture documentation, code repositories, and development artifacts from the application case; and focused feedback from architecture and domain experts.

6.6 Case and expert-panel selection

Application case: the entity that manages the Peruvian pension system is selected by convenience and relevance, given its status as an essential public service, its sensitivity in handling personal data, and its representativeness of critical State information systems. If access to this entity cannot be maintained during the design and evaluation phases, a comparable alternative organization will be identified (e.g., the Peruvian educational scholarship system), preserving the same level of criticality and regulatory constraints.

Expert panel: purposive sampling of 8 to 12 specialists in enterprise architecture, software engineering, and/or pension-system management, with at least three years of relevant experience, invited to the scenario-based evaluation.

6.7 Analysis plan

Quantitative: computation of precision, recall, defect escape rate, rework effort, and lead time for the two-agent pipeline and for the baseline, with descriptive comparison and, when the task sample size allows, non-parametric paired-data statistical tests (e.g., Wilcoxon), given the small sample size expected in a case study.

Qualitative: thematic analysis of the expert panel's feedback collected in the scenario-based evaluation, to identify patterns of perceived usefulness, feasibility, and risk.

6.8 Artifact prerequisites and restrictions

Prerequisites: alignment with Law No. 29733 (Protection of Personal Data), Law No. 31814, NTP-ISO/IEC 42001:2025, and Supreme Decree No. 029-2021-PCM (approving the Regulations of Legislative Decree No. 1412, the Digital Government Law, TITLE VIII, DIGITAL ARCHITECTURE); restriction of the artifact to development and verification (not production execution); data minimization with no identifiable personal pension data; and local / on-device processing with access limited to strictly necessary code and data.

7. Ethical Considerations

Personal data protection

The design complies with Law No. 29733 on the Protection of Personal Data: no identifiable personal data of pension-system members is collected or processed; all empirical evidence comes from technical records (logs, code, architectural documentation) and expert feedback, not from the data subjects of the pension system.

AI governance

The artifact is designed in conformance with Law No. 31814 (promoting the use of artificial intelligence) and NTP-ISO/IEC 42001:2025 (AI management systems), which includes traceability of the agents' decisions and the restriction of their scope to the development and verification phases.

Informed consent of the expert panel

Participation of experts in the scenario-based evaluation will be voluntary, preceded by informed consent, with the option to anonymize their responses and to withdraw from the study at any time without penalty.

Institutional confidentiality

Enterprise-architecture documentation, code repositories, and development artifacts provided by the collaborating entity will be handled under confidentiality agreements; the publication of results will avoid disclosing information that could compromise the security of the entity's systems.

Institutional ethics review

The protocol will be submitted to the university's research ethics committee and, where applicable, to the formal authorization of the collaborating public entity, before data collection and expert evaluation begin.

Risk management

- Risk of losing access to the collaborating entity: a comparable alternative organization is defined in advance (Section 6.6).
- Risk of correlated failures between the generator and verifier agents (if both are generative models, they could share training data, biases, and blind spots): mitigated by making the conformance layer at least partially deterministic (static analysis, policy as code); the observed degree of independence will be documented empirically.
- Risk of misuse of the artifact beyond its authorized scope: its operation is explicitly restricted to development/verification environments, never production.

8. Expected Results

- A documented reference-architecture artifact (architectural views, agent roles, conformance mechanism) for the governed development of critical State information systems.
- Comparative empirical evidence between the two-agent pipeline and the single-agent baseline, in terms of precision/recall, defect escape rate, rework effort, and lead time.
- Results of the scenario-based architecture evaluation, with experts' perceptions of usefulness, feasibility, and risk.

— Transferable design principles, in particular the conformance principle and the governed separation between generation and verification, applicable to other critical public-sector information systems.

— Empirical evidence on the degree of independence (or correlated failure) between generator and verifier agents, already identified in the Paradigm Justification Statement as a potentially publishable result in its own right.

— At least one publication in venues such as IEEE Software, Journal of Systems and Software, Government Information Quarterly, DESRIST, or the European Conference on Software Architecture (ECSA).

— Technical inputs that could inform AI-governance guidelines for software development in Peruvian public entities.

9. Timeline and Budget

The timeline is organized into six semesters (three years), corresponding to the six activities of the DSR cycle (Peffers et al., 2007) and to the feasibility limit identified in the Method Fit Matrix. Dates and amounts are indicative and will be adjusted to the doctoral program's funding rules.

Semester	DSR Activity	Key milestones
S1	Problem identification and motivation	Expanded literature review; Paradigm Justification Statement and Method Fit Matrix (completed).
S2	Definition of solution objectives	Architectural and regulatory requirements systematized; research protocol validated on OSF.
S3	Design and development	Specification of the reference architecture and conformance mechanism; prototype started.
S4	Design and development / Demonstration	Functional prototype of the generator–verifier pipeline; collaboration agreement with the case entity.
S5	Demonstration and evaluation	Execution of the application case; metric collection; scenario-based evaluation with experts.
S6	Evaluation and communication	Final analysis, design principles, manuscript writing, and research defense.

Indicative budget

Item	Rationale
Local computing infrastructure (workstation / GPU)	Local/on-device execution of the agents, consistent with the confidentiality requirement.
Licenses for static-analysis and policy-as-code tools	Support for the deterministic conformance-verification layer.
Incentives or symbolic compensation for the expert panel	Voluntary participation in the scenario-based evaluation.
Open-access publication fees (APC) and conference attendance	Dissemination in IEEE Software, JSS, GIQ, DESRIST, or ECSA.
Contingencies and coordination with the collaborating entity	Managing the risk of a change in access to the entity (Section 7).

Note: exact amounts per line item will depend on the doctoral program's funding scales and on the agreement with the collaborating entity; they are presented here as categories to be completed before formal submission to the Graduate School.

10. Bibliography

Academic references

Gregor, S., & Hevner, A. R. (2013). Positioning and Presenting Design Science Research for Maximum Impact. MIS Quarterly, 37(2), 337–355.

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design Science in Information Systems Research. MIS Quarterly, 28(1), 75–105.

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A Design Science Research Methodology for Information Systems Research. Journal of Management Information Systems, 24(3), 45–77.

Regulatory framework

Law No. 29733, Law on the Protection of Personal Data (Peru).

Law No. 31814, Law promoting the use of Artificial Intelligence for the country's economic and social development (Peru).

NTP-ISO/IEC 42001:2025, Artificial intelligence management systems — Requirements.

Supreme Decree No. 029-2021-PCM, approving the Regulations of Legislative Decree No. 1412, the Digital Government Law, TITLE VIII, DIGITAL ARCHITECTURE (Peru).

Note: this list will be expanded during the formal literature review (Section 4) with the specific sources on enterprise architecture, generative AI agents, and conformance verification identified in the systematic search.